SUN MON TUE WED THU FRI SAT

HW 1

1. $p(x_i \mid \mu, \Sigma) = \dfrac{1}{(2\pi)^{\frac{1}{2}d} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(x_i-\mu)^T \Sigma^{-1}(x_i-\mu)\right)$

$p(D \mid \mu, \Sigma) = \prod_{i=1}^{n} p(x_i \mid \mu, \Sigma)$, Likelihood function

log-likelihood:

$$l(\mu, \Sigma) = -\frac{nd}{2}\ln(2\pi) - \frac{n}{2}\ln|\Sigma| - \frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^T\Sigma^{-1}(x_i-\mu)$$

Set $\frac{\partial l}{\partial \mu} = 0$.

since for a symmetric matrix A:

$$\frac{\partial}{\partial z}(z-a)^T A (z-a) = 2A(z-a)$$

$$\frac{\partial l}{\partial \mu} = \sum_{i=1}^{n}\Sigma^{-1}(x_i-\mu) = 0$$

$$\Sigma^{-1}\left(\sum_{i=1}^{n} x_i - \sum_{i=1}^{n}\mu\right) = 0$$

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{n}\mu$$

$$\sum_{i=1}^{n} x_i = n\mu$$

$$\hat{\mu} = \frac{\sum_{i=1}^{n} x_i}{n} = \bar{x}$$

Let $W = \Sigma^{-1}$

$$\sum(x_i-\mu)^T W (x_i-\mu) = tr\left(W\sum(x_i-\mu)(x_i-\mu)^T\right)$$

Let $S_0 = \sum(x_i-\mu)(x_i-\mu)^T$

$$l(W) = \frac{n}{2}\ln|W| - \frac{1}{2}tr(WS_0)$$

$$\frac{\partial l}{\partial W} = \frac{n}{2}W^{-1} - \frac{1}{2}S_0^T = 0$$

$$nW^{-1} = S_0^T$$

$$n\Sigma = S_0^T$$

$$\hat{\Sigma} = \frac{S_0^T}{n} = \frac{S_0}{n}$$

$$\hat{\Sigma} = \frac{\sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T}{n}$$

2. $E[\hat{\mu}] = E\left[\frac{\sum x_i}{n}\right] = \frac{1}{n}\sum E[x_i] = \frac{n\mu}{n} = \mu$, So it's unbiased.

$$A = \sum_{i=1}^{n}(x_i-\bar{x})(x_i-\bar{x})^T = \sum_{i=1}^{n}[(x_i-\mu)-(\bar{x}-\mu)][(x_i-\mu)-(\bar{x}-\mu)]^T$$

$$= \sum_{i=1}^{n}(x_i-\mu)(x_i-\mu)^T - n(\bar{x}-\mu)(\bar{x}-\mu)^T$$

$$E[(x_i-\mu)(x_i-\mu)^T] = Var(x_i) = \Sigma$$

$$E[(\bar{x}-\mu)(\bar{x}-\mu)^T] = Var(\bar{x}) = \frac{\Sigma}{n}$$

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$E[A] = n\Sigma - n\left(\frac{\Sigma}{n}\right) = n\Sigma - \Sigma$

$E[\hat{\Sigma}] = \frac{n-1}{n}\Sigma$ so it's biased.

The unbiased estimator S is:

$$S = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^T$$

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